



ECE 205 SPRING 2026

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Homework 1

Submission method: Please submit .cpp source files and .pdf for output file on the canvas.

1. **(25 pts)** Write a program that will take as input from the console an integer that represents a length of time in seconds. The program will then output the number of hours, minutes, and seconds that corresponds to the user input. In addition, the program will output the number of hours as a real number and the number of minutes as a real number. For example, if the user enters 50391 seconds, then the program will first output 13 hours, 59 minutes, and 51 seconds. Then the program will output 13.9975 hours, and then output 839.8500 minutes. Use appropriate variable/constant names and types and format your output neatly. Format your real numbers to have 4 digits past the decimal point/radix. Test your code with several inputs to make sure the calculations are correct.

2. **(25 pts)** An employee is paid at the rate of \$36.75 per hour for regular hours (up to 40 hours/week). Any overtime hours are paid at a rate of 1.5× the base hourly rate. Taxes/dues are deducted from the worker's gross pay as follows: 6% for Social Security, 15% for federal income, 4% for state, 1% for Bethlehem local, and 20\$ in union dues. If the worker has 2 or more dependents, an additional 40\$ is deducted for health insurance. Write a program to read in the number of hours worked in a week and number of dependents from the console and output the worker's gross pay, all withholding amounts individually, and the net take-home pay for the week. Assume hours are entered as real numbers. Outputs do not have to contain the '\$' sign, but have to be output with 2 digits of precision. Use appropriate variable/constant names and types and format your output neatly. Test your code with several inputs to make sure the calculations are correct.

3. **(50 pts)** You have just purchased a stereo system that cost \$1,000 on the following (pretty bad) credit plan: no down payment, an interest rate of 18% per year (1.5% per month), and monthly payments of \$50. The monthly payment of \$50 is used to pay the interest first, and whatever is left is used to pay part of the remaining debt (similar to a mortgage). For example: the first month you pay 1.5% on \$1,000 or \$15 in interest and the rest (\$35) is deducted from your debt, which puts you at a debt of \$965.00. In the second month, your interest is \$14.48 (1.5% of \$965.00) and from your \$50 payment, only \$35.52 go towards the debt, leaving you with \$929.48 in debt and so on.

Write a program that will tell you how many months it will take you to pay off the loan, as well as the total amount of interest paid over the life of the loan and the overall amount paid. Use a loop to calculate the amount of interest and the size of the debt after each month. The last payment may be less than \$50 if the debt is small, but do not forget the interest. If you owe exactly \$50.00 for your last

month, then your monthly payment of \$50 will not pay off your debt, though it will come very close (one month's interest on \$50.00 is 75 cents).

Output current values for all variables that are part of the loan in a table as follows (use 2 decimal places for variables that need it and right-align each column):

Month #	Interest	Principal	Balance	Total Interest	Total Paid
1	\$15.00	\$35.00	\$965.00	\$15.00	\$50.00
2	\$14.48	\$35.52	\$929.48	\$29.48	\$100.00
etc...					